

SOURABH CHAUHAN

Software Engineer | GenAI Developer | AI/ML Automation Engineer
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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience designing and developing automation platforms, backend applications, and AI-powered solutions for banking, healthcare, and enterprise environments. Strong expertise in Python, SQL, FastAPI, GitHub Actions, Docker, Kubernetes, Linux, REST APIs, automated testing, and cloud-native development. Proven experience building production-grade automation frameworks, integrating AI-assisted development tools, optimizing software delivery pipelines, and delivering high-quality enterprise solutions in Agile environments.

ACADEMIC QUALIFICATIONS

Master of Science in Computer Science

Lakehead University, Canada | GPA: 4.0/4.0 | **Aug 2024 - Sept 2025**

Bachelor of Computer Applications

Amity University, India | GPA 3.7/4.0 | **Sept 2016 - Nov 2019**

PROFESSIONAL EXPERIENCE

AI Evaluation/Python Developer - Outlier AI (Freelance)

Nov 2025 - Jun 2026

Enterprise clients: Nokia Solutions and Networks

- Evaluated AI-generated responses using structured quality frameworks to improve model accuracy, reliability, consistency.
- Performed model performance analysis by identifying reasoning gaps, hallucinations, factual inaccuracies, formatting issues, and policy violations.
- Generated human feedback used for continuous model optimization, retraining, and alignment improvements.
- Developed an AI agent to automate C#-to-Python code conversion for Nokia Solutions' internal applications, ensuring functional equivalence, code compatibility, and seamless integration with existing systems.

Systems Engineer - Tata Consultancy Services

April 2023 - Aug 2024

Enterprise clients: Cigna Healthcare, Express Scripts & Lloyds Bank Group.

- Developed an AI-powered Chatbot that automated 8,000+ monthly IT support requests, reducing L1 support workload by 70% using Python, NLP pipelines, ServiceNow, REST APIs, and rule-based workflow orchestration.
- Engineered an AI Knowledge Assistant using Python, FastAPI, LangChain, and RAG, architecting secure document ingestion, intelligent chunking, vector embeddings, semantic retrieval, and LLM-powered question answering to deliver context-aware, grounded responses across enterprise knowledge while reducing manual information discovery.
- Introduced Dockerized microservices and GitHub Actions CI/CD pipelines, enabling repeatable deployments, Pytest-driven validation, and significantly improving release reliability across distributed services.
- Designed a centralized reporting and diagnostics platform that extracted, validated, and analyzed infrastructure health, patch compliance, and operational metrics, reducing review effort by 95% and improving data accuracy for operational decision-making.
- Collaborated with business stakeholders and infrastructure teams to translate operational requirements into scalable automation and reporting solutions, delivering projects on schedule while improving process efficiency.

Asst. Systems Engineer - Tata Consultancy Services

Feb 2022 - Mar 2023

- Developed machine learning pipeline using Python, XGBoost, and Logistic Regression models to classify incidents based on historical ticket data. Integrated predictions into ServiceNow to improve incident routing and reduce manual triage.
- Engineered AWS and PySpark ETL pipelines that extracted, transformed, validated, and consolidated large operational datasets into analytics-ready data models supporting enterprise reporting and performance analysis.
- Designed and implemented an MLOps platform using Python and Kafka to ingest and process infrastructure health events, applying Isolation Forest, and rule-based decisioning to detect anomalies and automatically route diagnostic alerts.
- Modernized insurance ETL pipelines using Python and PySpark to process data from ADLS and Power BI dashboards for customer, policy, retention, and claims insights.
- Delivered 12+ enterprise automation solutions across the full SDLC—from requirements analysis and architecture through deployment and production support—saving over 150 engineering hours of manual effort per month.

Automation Engineer - Tata Consultancy Services

Mar 2020 - Jan 2022

- Developed automation solutions that orchestrated VMware provisioning, infrastructure validation, and configuration management, reducing provisioning time from 3 hours to under 10 minutes.
- Automated Linux application and Modules deployments using Python, Bash, Ansible, and CI/CD workflows, ServiceNow, eliminating 98% of manual release activities across enterprise environments.
- Improved Windows patch deployment throughput by 70% by replacing manual SCCM validation activities with Python automation that verified deployment status, compliance, and remediation before production rollout.
- Streamlined infrastructure maintenance by automating user provisioning, snapshot lifecycle management, log rotation, and operational tasks, reducing repetitive engineering effort by 80%.

- Developed backend modules for enterprise HRMS and Home Services platforms using C#, .NET, and SQL Server, implementing secure role-based access control, workflow automation, and database-driven business logic.
- Implemented modules for employee access management, service booking workflows, and user-role authorization systems.
- Supported application testing, defect fixes, database updates, and technical documentation during development and release cycles.

CORE SKILLS

- **Programming:** Python, Java, SQL, PowerShell, C#, Bash, YAML
- **Backend & APIs:** FastAPI, Flask, Django, REST APIs, Microservices, Object-Oriented Programming
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, CI/CD, GitHub Actions, Ansible, Linux
- **Databases & Data Engineering:** MySQL, PostgreSQL, Oracle Database, MongoDB, SQLite, Databricks, PySpark, ELT/ETL Pipelines
- **AI & Machine Learning:** LLM, Scikit-learn, TensorFlow, PyTorch, NLP, RAG, LangChain, LangGraph, AI Agents, Prompt Engineering, AI Automations, Ollama, Hugging face, LangSmith
- **Testing:** Pytest, Unit Testing, Integration Testing, API Testing, Selenium, A/B Testing, Production Support
- **Tools & Practices:** Git, Jira, Confluence, ServiceNow, Agile, Scrum, ITIL

AWARDS & RECOGNITION

- **Star of the Month Award** — Tata Consultancy Services
- **On-the-Spot Team Excellence Award** for enterprise automation delivery
- **Client Recognition Awards** for improving operational efficiency across enterprise environments.
- **Special Initiative and Contextual Master Awards** for introducing AI/ML implementation and CI/CD pipeline improvements for client delivery teams

CERTIFICATIONS

- Oracle Certified Generative AI Professional [Link](#)
- Certified Data Science Professional [Link](#)
- Oracle Certified DevOps Professional [Link](#)
- IBM Data Analysis with Python | [Link](#)
- MongoDB for SQL Experts | [Link](#)
- LangChain Academy: LangChain – Python | [Link](#)